Dovari Rahul

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SUMMARY

Detail-oriented Computer Science student with practical experience in machine learning, natural language processing, computer vision, and full-stack development. Proficient in building AI-powered systems using TensorFlow, Scikit-Learn, NLTK, and Sentence Transformers. Experienced in developing end-to-end ML pipelines, semantic search systems, and deploying models through Flask APIs. Seeking internship opportunities to apply AI and software engineering skills in real-world projects.

EDUCATION

NRI Institute of Technology, Agiripalli, Andhra Pradesh Aug. 2023 – May 2027 (Expected)
Bachelor of Technology in Computer Science and Engineering

The Hindu College, Machilipatnam, Andhra Pradesh June 2021 – May 2023
Intermediate Education, MPC (Maths, Physics, Chemistry)

EXPERIENCE

Data Analyst Intern | Bluestock 2025
Python, Pandas, Excel, Stock Market Analysis

- Analyzed historical and live price data of Nifty 50 index stocks to identify trends, volatility patterns, and trading signals.
- Built data pipelines and visualizations using Python and Excel to summarize sector-wise performance and support investment insights.

College Website for Faculty Patents & Books | NRI Institute of Technology 2025
HTML, CSS, JavaScript, Flask, MySQL

- Developed a college web application to manage and track faculty patents, publications, and books in a centralized database.
- Enabled faculty to submit and update records while allowing admins to verify entries and generate department-wise reports.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, HTML5, CSS3
Machine Learning & AI: TensorFlow, Keras, Scikit-Learn, PyTorch, NLTK, Sentence Transformers, OpenCV
Data Science: Pandas, NumPy, Matplotlib, Seaborn, Feature Engineering
NLP & Information Retrieval: FAISS, Semantic Search, Text Processing, Embeddings
Web Frameworks: Flask, FastAPI, REST APIs
Database: MySQL, Vector Databases
Developer Tools: Git, GitHub, VS Code, Jupyter Notebook, Google Colab
Cloud & AI Services: Google Gemini API

PROJECTS

AI Resume Screening System | Python, Flask, NLTK, FAISS, Sentence Transformers, Gemini API Feb. 2026 – present

- Engineered an AI-powered recruitment system automating candidate screening using NLP and semantic search, processing 200+ technical and soft skills extraction from resumes.
- Implemented multi-modal resume parsing supporting PDF and DOCX formats using PyPDF2, pdfplumber, and python-docx libraries for robust text extraction.
- Built a semantic matching engine using Sentence Transformers (all-MiniLM-L6-v2) generating 384-dimensional embeddings indexed in FAISS vector database for scalable search across thousands of resumes.
- Developed a composite ranking algorithm combining semantic similarity (50%), TF-IDF matching (20%), and skill overlap (30%), achieving 85%+ relevance accuracy in candidate ranking.
- Integrated Google Gemini API for AI content detection and ATS evaluation, providing comprehensive resume scoring across 4 critical criteria.
- Designed a premium glassmorphism dark-themed web UI using Flask backend with RESTful API endpoints for resume upload, matching, and analytics.

AI-Powered Live Emotion Analytics System | Python, TensorFlow, OpenCV, Scikit-Learn Nov. 2025 – Dec. 2025

- Designed a real-time emotion recognition system using Convolutional Neural Networks (CNN) for facial expression classification across 7 emotion states.
- Implemented a computer vision pipeline processing live camera feeds and video streams using OpenCV with Haar Cascade face detection achieving 30+ FPS processing speed.
- Performed advanced image preprocessing including normalization, data augmentation, and feature extraction techniques, improving model accuracy by 15%.
- Trained a deep learning model on 35,000+ labeled facial images achieving 87% validation accuracy with dropout regularization to prevent overfitting.
- Built an interactive real-time analytics dashboard visualizing emotion distributions and trends using Matplotlib for live monitoring applications.

AI-Powered Speech Recognition & Synthesis System | Python, Whisper, Librosa, pyttsx3, FastAPI June 2025 – Sep. 2025

- Developed an end-to-end Speech-to-Text (STT) system using OpenAI's Whisper acoustic model, processing audio files with 95%+ transcription accuracy.
- Implemented an offline Text-to-Speech (TTS) engine using pyttsx3 with customizable voice parameters supporting multiple languages and speech rates.
- Designed a complete ML pipeline from raw audio waveform processing to feature extraction (MFCC, Mel Spectrogram) using Librosa, through model inference and decoding.
- Optimized the model inference pipeline, reducing processing latency by 4× through efficient batching, caching mechanisms, and NumPy vectorization.
- Deployed the system as a RESTful API using FastAPI with automatic documentation and asynchronous request handling for scalable speech processing.

CERTIFICATIONS

Professional Development – 2025-26

- Generative AI (GenAI) Certification – Infosys
- Machine Learning Certification – Kaggle
- Machine Learning & AI Foundation
- AI for Business Profession – HP
- Hackathon Participation Certificate – PSCMR

ADDITIONAL INFORMATION

Languages : English (Professional), Telugu (Native)

Interests: Open Source Contribution, AI Research, Competitive Programming, Technical Writing